

Corrosion behavior of FeCoNiCrCu_x high-entropy alloys in 3.5% sodium chloride solution

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Abstract

The corrosion behavior of FeCoNiCrCu_x high-entropy alloys in 3.5% NaCl solution is reported in this paper. The corrosion resistance has been evaluated using immersion tests and potentiodynamic polarization measurements. Our results show that the increase of copper content in FeCoNiCrCu_x alloys also increases the tendency to localized corrosion of these alloys. This is due to Cu-rich interdendrite and Cu-depleted (Cr-rich) dendrite in these alloys set up a active/noble cell of appreciable potential difference, and the galvanic action results in preferentially attack along the interdendrites of copper containing high-entropy alloys.

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1. Introduction

In the past, the design concept of new alloy systems has been conventionally based on an element as their major constituent, such as Fe-based and Ni-based alloy systems, and other lesser elements as their constituents for the enhancement of properties and performances. This design concept has also restricted the development of new materials in recent years. To break through this cognitive concept of alloy design, a new alloy system was developed [1,2]. This new concept of alloy system is also called “high-entropy alloy”. High-entropy alloys have multielements as their major constituents, and the contents of each major constituent is greater than 5 at.% and less than 35 at.%. Because there is no matrix element in high-entropy alloy system, each constituent element in this system could be regarded as a solute atom. To increase the entropy, each major element of these alloy systems is in equimolar or near-equimolar ratio. According to the previous researches [1–6], these high-entropy alloy systems have many good properties, such as ease of amorphization, high hardness and strength, good thermal stability. However,

the corrosion data of high-entropy alloys in most aqueous solutions are not available. The purpose of this study is to investigate the immersion behavior and electrochemical polarization behavior of FeCoNiCrCu_x high-entropy alloys in 3.5% NaCl solution.

2. Experimental methods

2.1. Materials preparation

The alloys used in this study were FeCoNiCr ($x=0$), FeCoNiCrCu_{0.5} ($x=0.5$) and FeCoNiCrCu ($x=1$). These alloys were melted by arc melting from a mixture of high purity (99.99%) of Fe, Co, Ni, Cr, and Cu metals under a titanium-gettered atmosphere of argon. For comparison, AISI 304L stainless steel was also used in this study.

The composition analyses of elements within these alloys were carried out by scanning electron microscope (SEM) equipped with an energy dispersive X-ray spectrometry (EDX). X-ray diffraction (XRD) using Cu K α radiation was used to identify crystalline phases and obtain the lattice parameters of these alloys. The XRD investigation was per-

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formed with the diffraction angle (2θ) from 20° to 100° at a speed of 1° min^{-1} .

2.2. Immersion tests

Immersion tests were conducted in aerated 3.5% NaCl solution at 25°C for 30 days. To remove adherent corrosion products after immersion tests, the corroded specimens were cleaned in 10% HCl solution with 1 g L^{-1} hexamethylenetetramine followed by distilled water and acetone. Then, the specimens were dried in an oven for 1 h and weighed to 0.1 mg. The corrosion rate then can be calculated from the weight loss. The surface morphology of each specimen was also examined by SEM to determine the extent of degradation and to identify the corrosion mechanisms.

2.3. Potentiodynamic polarization measurements

Potentiodynamic polarization measurements were carried out in a typical three-electrode cell setup with the specimen as a working electrode, a saturated calomel reference electrode (SCE), and a platinum counter electrode. The potential of the working electrode was measured through the Luggin probe with respect to the reference electrode which was placed as close as possible to the specimen. The electrochemical polarization measurements were conducted in aerated 3.5% NaCl solution at 25°C under atmospheric pressure. The specimen was scanned potentiodynamically at a rate of 1 mV s^{-1} from the initial potential of -250 mV versus open circuit potential to the final potential of $1.6 \text{ V}_{\text{SCE}}$. Calibration was conducted in accordance with ASTM standard G5 [7] and G59 [8].

3. Results and discussion

3.1. Microstructure and phase constitution of as-cast FeCoNiCrCu_x alloys

The chemical compositions of specimens are listed in Table 1. The optical micrographs of these alloys are shown in Fig. 1. Fig. 2 illustrates the XRD patterns of as-cast FeCoNiCr, FeCoNiCrCu_{0.5}, and FeCoNiCrCu alloys. In this study, all these alloys basically belong to fcc structure. The small peaks slightly on the left of the matrix fcc of FeCoNiCrCu in XRD patterns is identified as Cu-rich interdendrites, which exhibits the same location of copper and have been

Table 1
EDX analyses of the compositions of FeCoNiCr, FeCoNiCrCu_{0.5}, and FeCoNiCrCu

	Alloys (wt.%)				
	Fe	Cr	Ni	Co	Cu
FeCoNiCr	24.58	25.12	25.92	24.90	–
FeCoNiCrCu _{0.5}	20.47	16.48	25.48	23.99	13.49
FeCoNiCrCu	18.78	16.25	21.78	21.50	21.69

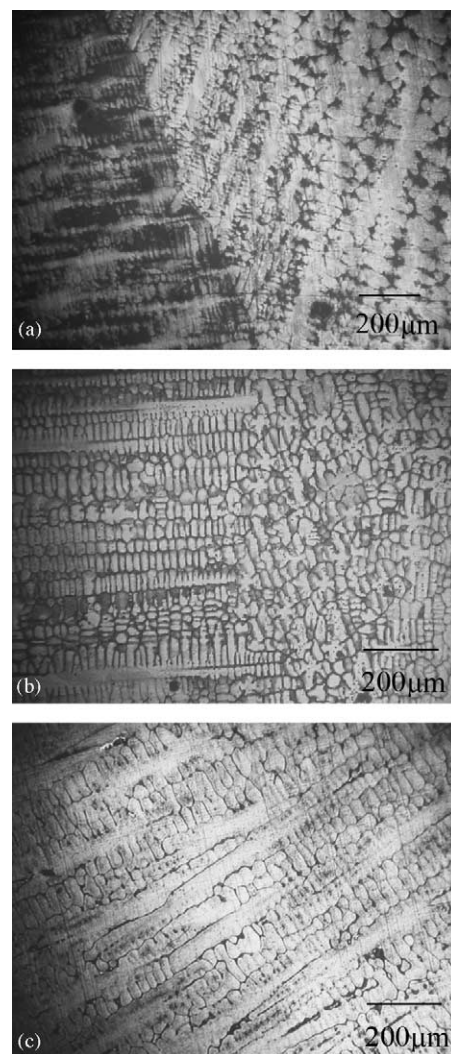


Fig. 1. Microstructure of as-cast FeCoNiCrCu_x alloys: (a) $x = 0$, (b) $x = 0.5$, and (c) $x = 1$.

reported previously [6]. As the Cu content decreases in these alloys, the peak intensities also changes due to the lattice distortion caused by different atomic size and still remains in fcc structure [6,9]. There is also no tendency to form complex or multiple phases when increasing the number of constituent elements. Because the wavelength of X-ray is constant ($\lambda = 1.54 \times 10^{-10} \text{ m}$), from the angle (θ) and plane (hkl) of each diffraction peak, a set of lattice parameter can be calculated by Bragg's law. These values of lattice parameter were plotted against $\cos^2\theta$, and the precise parameter of alloy (a_0) can be estimated by least squares and extrapolation method [10]. The lattice parameters of FeCoNiCr, FeCoNiCrCu_{0.5}, and FeCoNiCrCu in this study are 3.587×10^{-10} , 3.585×10^{-10} , and $3.582 \times 10^{-10} \text{ m}$, respectively.

Fig. 3 shows the SEM-backscattered electron images (BEI) of as-cast FeCoNiCrCu_{0.5} and FeCoNiCrCu alloys, which are composed of dendrites and interdendrites. According to the EDX analyses as listed in Table 2, the interdendrite (lighter phase) of as-cast FeCoNiCrCu_x alloy is Cu-

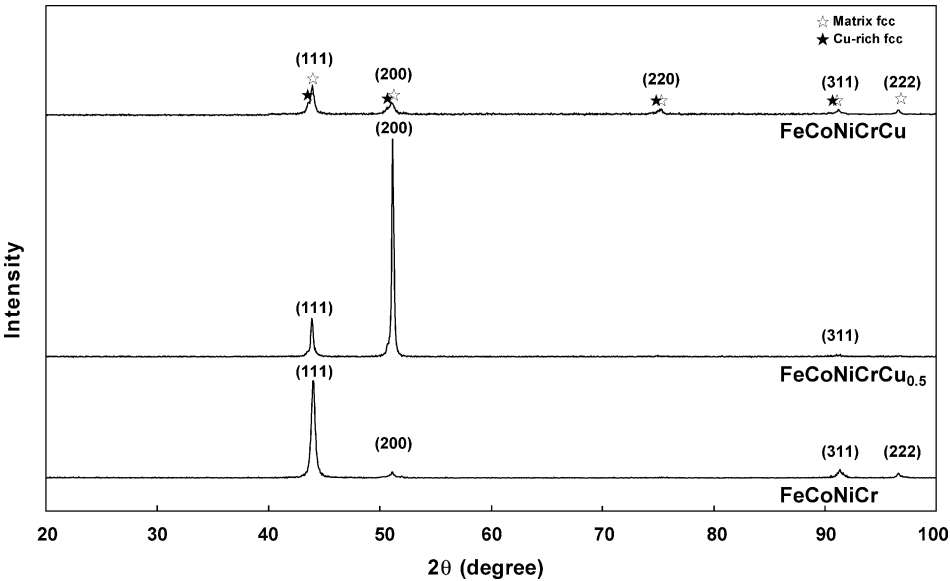


Fig. 2. XRD patterns of as-cast FeCoNiCrCu_x alloys.

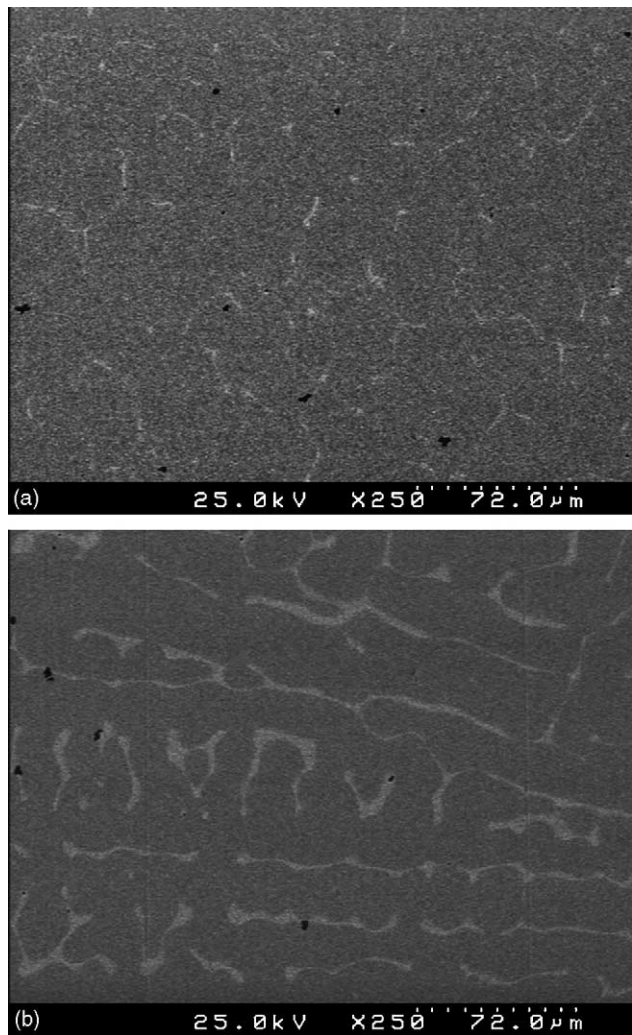


Fig. 3. Micrographs of as-cast FeCoNiCrCu_x alloys: (a) $x = 0.5$, and (b) $x = 1$.

rich phase, and dendrite (darker phase) is Cu-depleted phase. When decreasing the Cu concentration, the phenomenon of Cu segregation in interdendrites will be gradually removed. As listed in Table 3 [5,11–13], Cu element has weaker binding force (chemical mixing enthalpy, ΔH^{chem}) to other constituent elements than Fe, Co, Ni, and Cr. As a result of dendritic solidification, Cu solute atoms tend to segregate across the interdendrites and grain boundaries during the period of solidification. According to some previous studies, it is not easy to eliminate this phenomenon of Cu segregation by conventional heat treatment [5].

Table 2
EDX analyses of the dendrite and interdendrite structure of as-cast FeCoNiCrCu_{0.5} and FeCoNiCrCu alloys

	FeCoNiCrCu _{0.5} (at.%)		FeCoNiCrCu (at.%)	
	Interdendrite	Dendrite	Interdendrite	Dendrite
Cu	82.18	11.68	86.68	9.56
Fe	0.90	21.80	3.46	23.58
Co	1.08	24.45	2.67	24.67
Ni	11.60	22.73	5.41	21.34
Cr	4.24	19.34	1.78	21.08

Table 3
Chemical mixing enthalpy (ΔH^{chem}) of a pair of atoms [5,11–13]

	ΔH^{chem} (kJ mol ^{−1})
Cu–Fe	13
Cu–Co	10
Cu–Ni	4
Co–Ni	0
Cu–Cr	−1
Fe–Co	−1
Fe–Cr	−1
Fe–Ni	−2
Co–Cr	−4
Ni–Cr	−7

3.2. Properties of as-cast FeCoNiCrCu_x alloys

3.2.1. Immersion tests

The average corrosion rates of as-cast FeCoNiCrCu_x alloys were determined by using the weight loss method in 3.5% NaCl solution for 30 days immersion tests at 25 °C. The average corrosion rate can be calculated from [14]:

$$\text{corrosion rate (mm year}^{-1}\text{)} = \frac{8.76 \times 10^4 \times W}{T \times A \times D} \quad (1)$$

where W is the total weight loss (in g) after exposure time T (in h), D is the density of alloy (in g cm⁻³), and A (in cm²) is the area of the specimen. The D values of FeCoNiCr, FeCoNiCrCu_{0.5}, and FeCoNiCrCu are 8.25, 8.38, and 8.36, respectively.

The average corrosion rates in 3.5% NaCl solution for 30 days immersion test at 25 °C are summarized in Table 4. In the FeCoNiCrCu_x alloys, the Cu addition significantly increases the corrosion rates and leads to interior localized corrosion in 3.5% NaCl solution. Cu alloying element has

Table 4

Average corrosion rates (mm year⁻¹) of as-cast FeCoNiCrCu_x alloys in 3.5% NaCl solution at 25 °C

	FeCoNiCr	FeCoNiCrCu _{0.5}	FeCoNiCrCu
Immersion test ^a	7.62×10^{-4}	8.89×10^{-3}	1.14×10^{-2}
Polarization curve ^b	3.31×10^{-4}	7.46×10^{-3}	1.37×10^{-2}

^a Average corrosion rate is converted from weight loss data.

^b Average corrosion rate is obtained from Tafel region extrapolation and Faraday's law.

negative effect on the corrosion resistance of FeCoNiCrCu_x alloys.

The surface appearances of as-cast FeCoNiCrCu_x alloys are shown as Fig. 4. According to the SEM images, surface examination showed that the major types of corrosion were mainly localized corrosion and pitting, not uniform corrosion. Therefore, the corrosion rate converted from weight loss data could only be used as an index of extent of corrosion attack. Therefore, the order of weight loss due to localized corrosion and pitting for these alloys was FeCoNiCrCu > FeCoNiCrCu_{0.5} > FeCoNiCr.

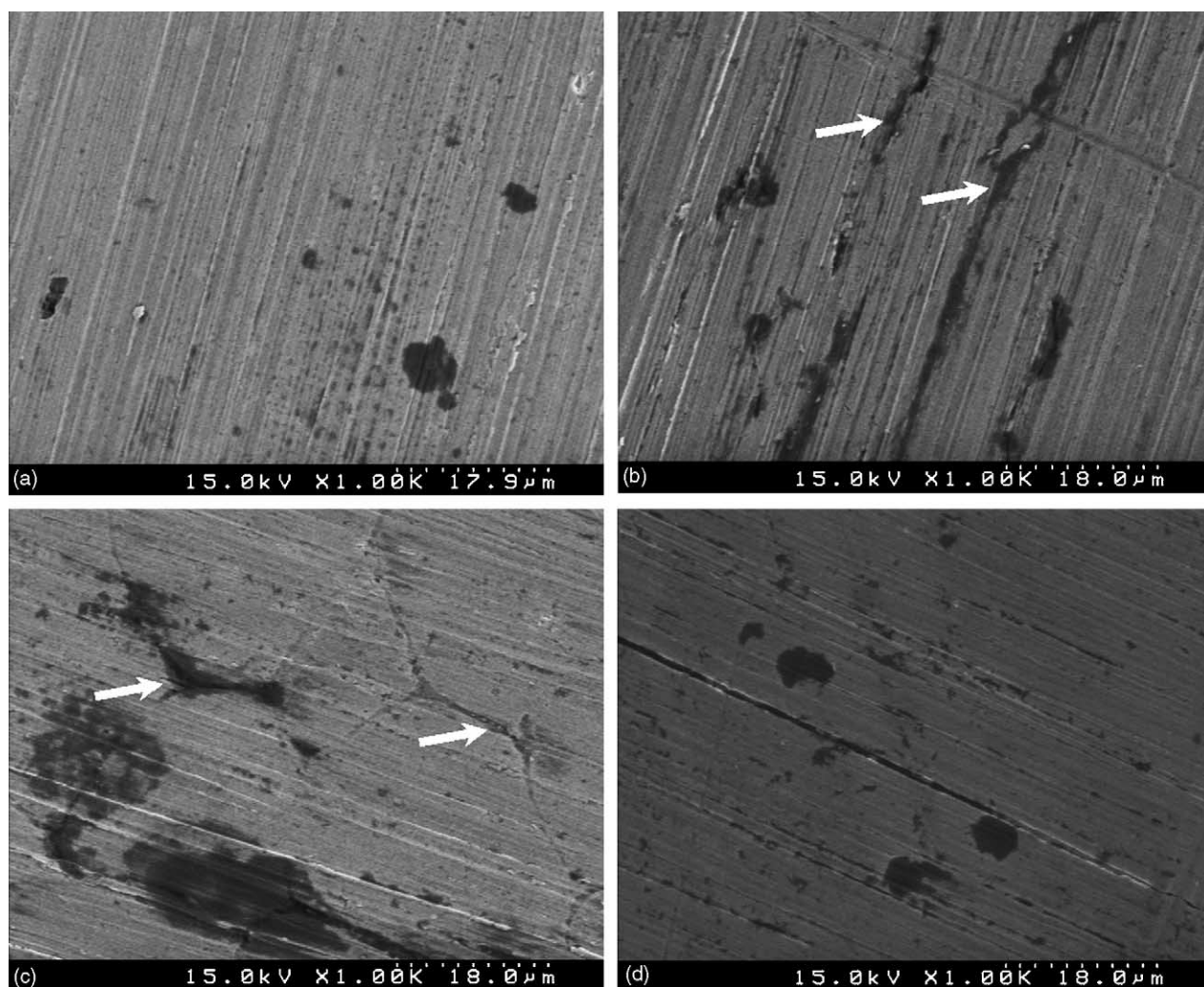


Fig. 4. Surface appearances of alloys after immersion tests in 3.5% NaCl solution for 30 days: (a) FeCoNiCr, (b) FeCoNiCrCu_{0.5}, (c) FeCoNiCrCu, and (d) 304L (white arrows to the localized corrosion causing by electrochemical attack along the interdendrites).

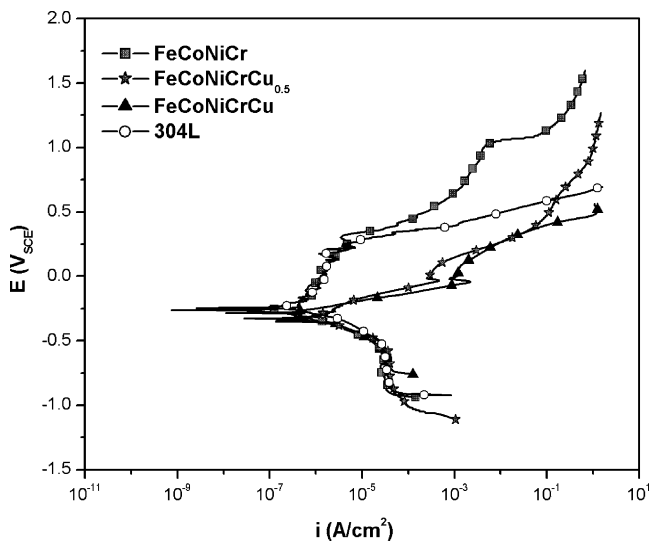


Fig. 5. Polarization curves for as-cast FeCoNiCrCu_x alloys and 304L in 3.5% NaCl solution at 25 °C.

The corroded surfaces of FeCoNiCr alloy and 304L stainless steel in 3.5% NaCl solution showed numerous small irregular shapes of dissolution pits. But the corroded surfaces of FeCoNiCrCu_{0.5} and FeCoNiCrCu alloys showed that the Cu-rich phases in interdendrites corrode preferentially. Cu alloying elements exert harmful effects on FeCoNiCrCu_x alloys that the corrosion resistance is degenerated, and damage is accelerated with an increase in the Cu content. According to Table 2, the interdendrites are also Cr-depleted zone and in active state. The active interdendrites are coupled to the Cr-rich and Cu-depleted passive dendrites. The low anode/cathode surface area ratio results in galvanic corrosion at the interdendrites, and the electrochemical action causes attack along the interdendrites in copper containing alloys.

3.2.2. Potentiodynamic polarization measurements

The potentiodynamic polarization curves of as-cast FeCoNiCrCu_x alloys in 3.5% NaCl solution are given in Fig. 5. The curve for 304L stainless steel is also plotted for comparison. Only limited regions of passivation are presented for the FeCoNiCrCu_{0.5} and FeCoNiCrCu alloys, which are much narrower than FeCoNiCr alloy and 304L stainless steel. The average corrosion rates obtained from electrochemical polarization measurements can be calculated from [14]:

$$\text{corrosion rate (mm year}^{-1}\text{)} = \frac{3.27 \times 10^{-3} \times i_{\text{corr}} \times \text{EW}}{D} \quad (2)$$

where EW is the equivalent weight of alloy, D is the density of alloy (in g cm⁻³), and i_{corr} (in $\mu\text{A cm}^{-2}$) is the corrosion current density of the specimen. The EW values of FeCoNiCr, FeCoNiCrCu_{0.5}, and FeCoNiCrCu are 26.56, 26.38,

Table 5

Electrochemical parameters of as-cast FeCoNiCrCu_x alloys and 304L stainless steel in 3.5% NaCl solution at 25 °C

	E_{corr}^a (V _{SCE})	E_{pit}^b (V _{SCE})	i_{corr}^c (A cm ⁻²)
FeCoNiCr	-0.26	0.31	3.15×10^{-8}
FeCoNiCrCu _{0.5}	-0.29	0.09	7.23×10^{-7}
FeCoNiCrCu	-0.33	0.08	1.32×10^{-6}
304L	-0.25	0.23	6.01×10^{-7}

^a E_{corr} : corrosion potential.

^b E_{pit} : pitting potential.

^c i_{corr} : corrosion current density.

and 26.56, respectively. The corrosion rates presented in Table 4 delineates that the results obtained from immersion test and polarization study are in correspondence with each other.

The electrochemical parameters of as-cast FeCoNiCrCu_x alloys and 304L stainless steel in 3.5% NaCl solution are also given in Table 5. The corrosion potentials for as-cast FeCoNiCrCu_x alloys vary from -0.26 to -0.33 V_{SCE}, and the pitting potentials vary from 0.08 to 0.31 V_{SCE}. FeCoNiCr alloy and 304L stainless steel have higher corrosion potential and pitting potential among all specimens. Because of Cu segregation in interdendrites and then causing non-protective passive film on surface, the passivation region of FeCoNiCrCu_x alloys with higher Cu addition is narrower than those without Cu addition in 3.5% NaCl solution. The larger passivation current was also observed in those alloys with Cu content. These indicate that FeCoNiCr alloy is easier to passivate and more corrosion-resistant in 3.5% NaCl solution and even shows better corrosion-resistant than 304L stainless steel.

4. Conclusions

1. The as-cast FeCoNiCrCu_x alloys ($x=0, 0.5$, and 1) belong to fcc structure. But for as-cast FeCoNiCrCu_{0.5} and FeCoNiCrCu alloys, the phenomenon of Cu segregation due to weaker bonding force to other elements is easily formed in interdendrites.
2. After the immersion tests for 30 days in 3.5% NaCl solution, the surface examination of as-cast FeCoNiCrCu_{0.5} and FeCoNiCrCu alloys showed that the major type of corrosion was galvanic action between interdendrite and dendrite, and then resulted in localized corrosion. The corroded surfaces of as-cast FeCoNiCr alloy and 304L stainless steel showed numerous small irregular shapes of dissolution pits. The order of weight loss due to localized corrosion and pitting for these alloys was FeCoNiCrCu > FeCoNiCrCu_{0.5} > FeCoNiCr. The corrosion damage is accelerated with an increase in the Cu content.
3. FeCoNiCr alloy is easier to passivate and more corrosion-resistant than FeCoNiCrCu_{0.5} and FeCoNiCrCu alloys in 3.5% NaCl solution. This phenomenon is caused by Cu segregation in interdendrites.

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